



DATA-500 PROJECT REPORT

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Abstract:

In this report we will look at many patient attributes with the goal of identifying characteristics patients with Alzheimer's are most likely to have. Identifying Alzheimer's early offers many benefits to not only the patient but the family of the diagnosed. Using Exploratory Data Analysis, we were able to find these characteristics to give patients and people at risk of Alzheimer's advanced notice of their potential to develop the condition.

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Introduction

Alzheimer's is the most common form of dementia². It causes problems with memory, thinking and behavior². Nearly 7 million Americans are living with Alzheimer's and 55 million worldwide are living with Alzheimer's and other dementias². Identifying it early can have many benefits such as medical benefits, emotional and social benefits, cost savings, and gives more time to prepare for the future².

The goal of this project is to identify and flag patients that are likely to have Alzheimer's currently or in the future to give them as much time as possible to do what they deem necessary.

Data Description

This dataset was found on Kaggle¹. The data is synthetically generated and contains 2,149 rows and 35 columns where each row represents a unique patient and each column containing demographic, lifestyle, and medical, details about the patient. There are no missing values in this dataset.

Column name	Type	Description
PatientID	Numeric	Unique identifier assigned to each patient (4751-6900)
Age	Numeric	The age of the patients (60-90)
Gender	Numeric	0-male, 1-Female
Ethnicity	Numeric	0-Caucasian, 1-African American, 2-Asian, 3-Other
EducationLevel	Numeric	0-None, 1-High School, 2-Bachelor's, 3-Higher
BMI	Numeric	Body mass of the patients (15-40)
Smoking	Numeric	0-No, 1-Yes
AlcoholConsumption	Numeric	Weekly consumption in units (0-20)
PhysicalActivity	Numeric	Weekly physical activity in hours (0-10)
DietQuality	Numeric	Diet quality score (0-10)
SleepQuality	Numeric	Sleep quality score (4-10)
FamilyHistoryAlzheimers	Numeric	0-Family history, 1-No History
CardiovascularDisease	Numeric	0-No, 1-Yes
Diabetes	Numeric	0-No, 1-Yes
Depression	Numeric	0-No, 1-Yes
HeadInjury	Numeric	0-No, 1-Yes
Hypertension	Numeric	0-No, 1-Yes
SystolicBP	Numeric	Systolic blood pressure (90-180) mmHg
DiastolicBP	Numeric	Diastolic blood pressure (60-120) mmHg
CholesterolTotal	Numeric	Total cholesterol levels (150-300)mg/dL
CholesterolLDL	Numeric	Low-density lipoprotein cholesterol levels (50-200) mg/dL

CholesterolHDL	Numeric	High-density lipoprotein cholesterol levels (20-100) mg/dL
CholesterolTriglycerides	Numeric	Triglycerides levels (50-400) mg/dL
MMSE	Numeric	Mini-Mental State Examination score (0-30). Lower scores indicate cognitive impairment.
FunctionalAssessment	Numeric	Functional assessment score (0-10). Lower scores indicate greater impairment.
MemoryComplaints	Numeric	0-No, 1-Yes
BehavioralProblems	Numeric	0-No, 1-Yes
ADL	Numeric	Activities of Daily Living score (0-10). Lower scores indicate greater impairment.
Confusion	Numeric	0-No, 1-Yes
Disorientation	Numeric	0-No, 1-Yes
PersonalityChanges	Numeric	0-No, 1-Yes
DifficultyCompletingTasks	Numeric	0-No, 1-Yes
Forgetfulness	Numeric	0-No, 1-Yes
Diagnosis	Numeric	0-No Alzheimer's, 1-Alzheimer's
DoctorInCharge	Character	Contains confidential information about the doctor in charge, "XXXConfid" is the value for all patients.

Exploratory Data Analysis

To get a better understanding of our dataset we created multiple plots and histograms to find correlations/trends in the data that can help us find commonalities among patients diagnosed with Alzheimer's. We focused on searching for differences in distributions that would lead us to believe there is an apparent impact that variable would have on diagnosis.

Figure 1

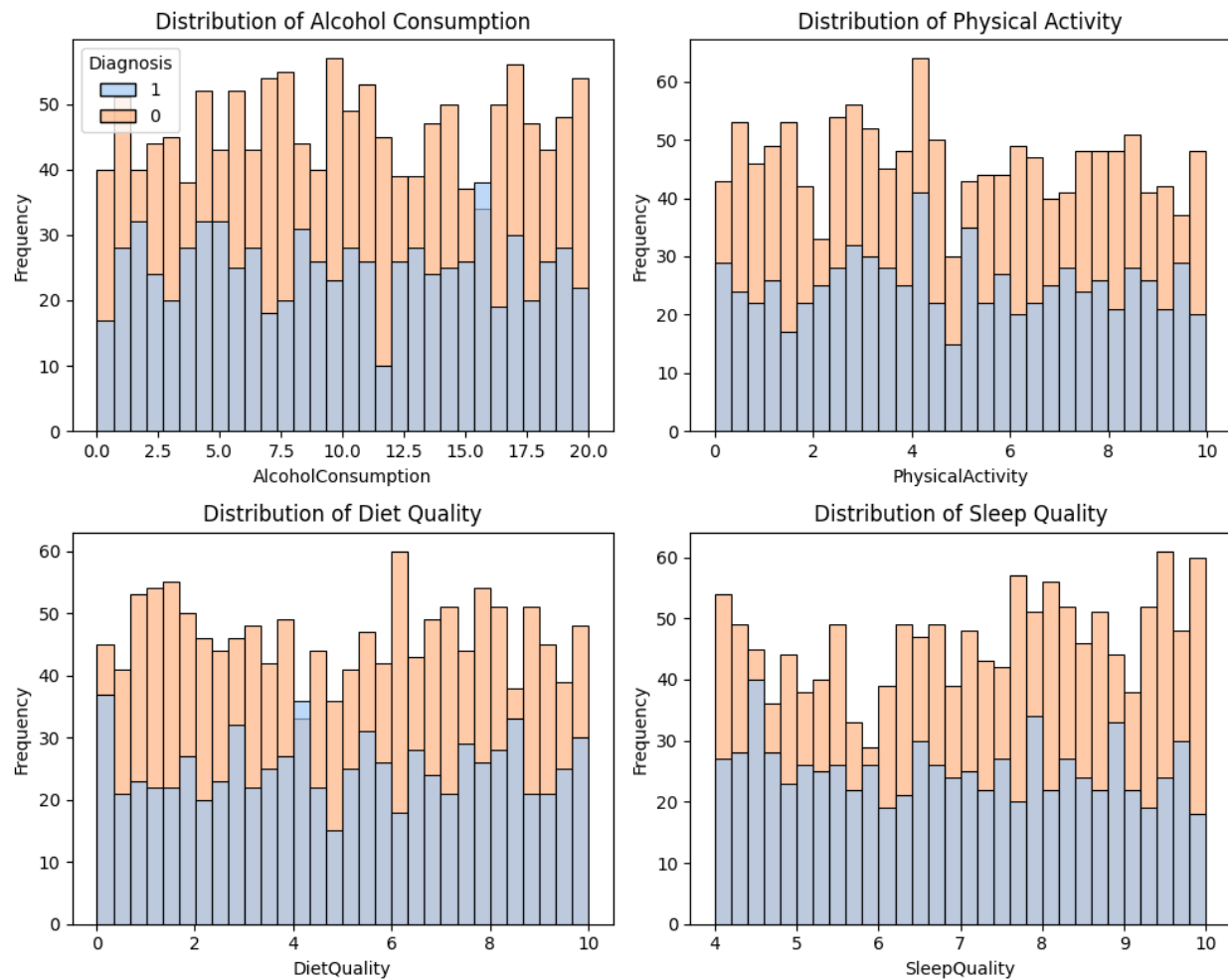


Figure 1: The histograms to the left show the distribution of various lifestyle habits of the patients in this dataset. They all share similar distribution shapes and a similar frequency of diagnosis throughout the variable range.

Figure 2

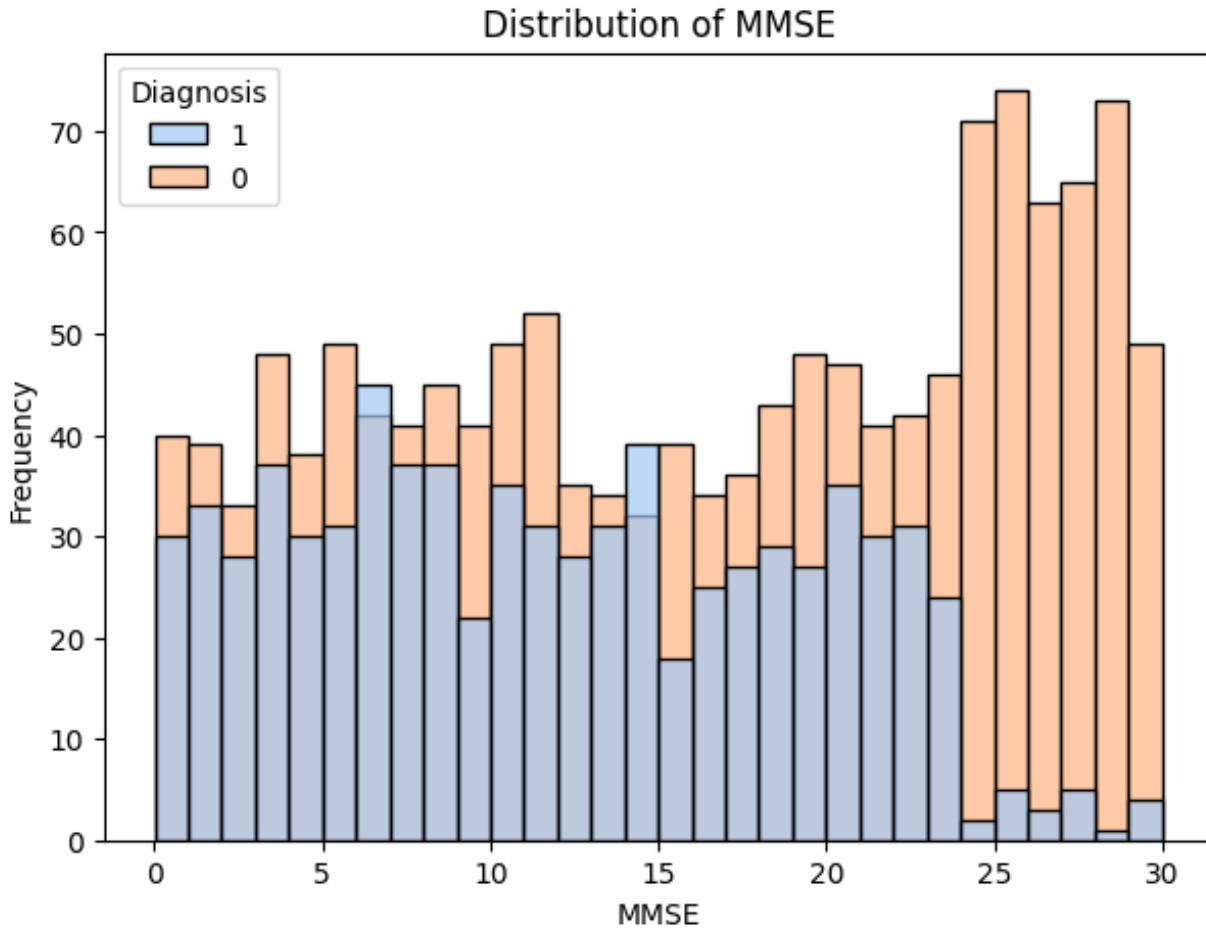


Figure 2: The histogram to the left shows the distribution of Mini-Mental State Examination score. Very few patients diagnosed with Alzheimer's scored above a 25. Lower scores indicate cognitive impairment so it would make sense fewer positively diagnosed patients would score high on the MMSE.

Figure 3

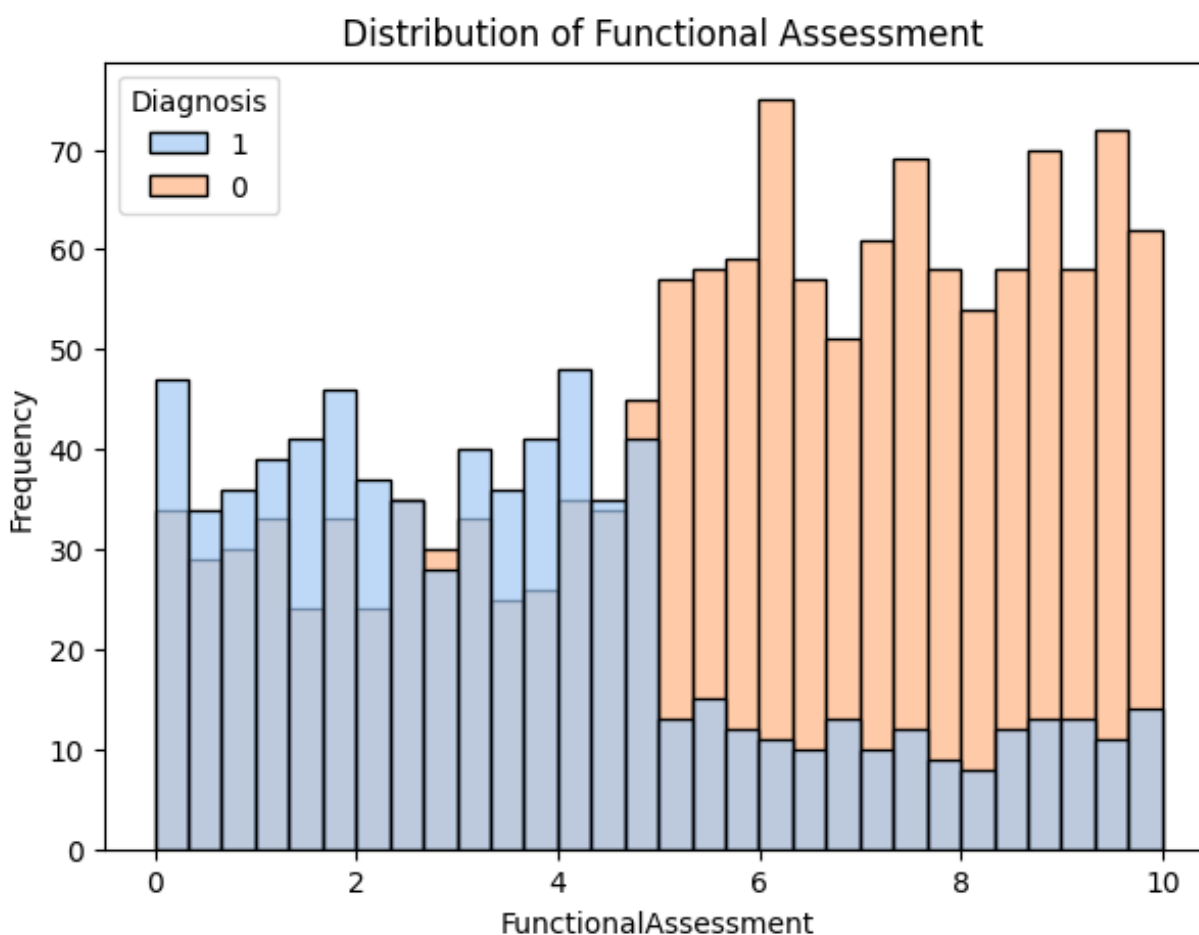


Figure 3: The histogram to the left shows the distribution of Functional Assessment score. Similar to Figure 2, Lower scores indicate greater impairment. The distribution of the positive case has a right skew while the distribution of the negative case has a left skew showing more positively diagnosed patients show greater impairment.

Figure 4

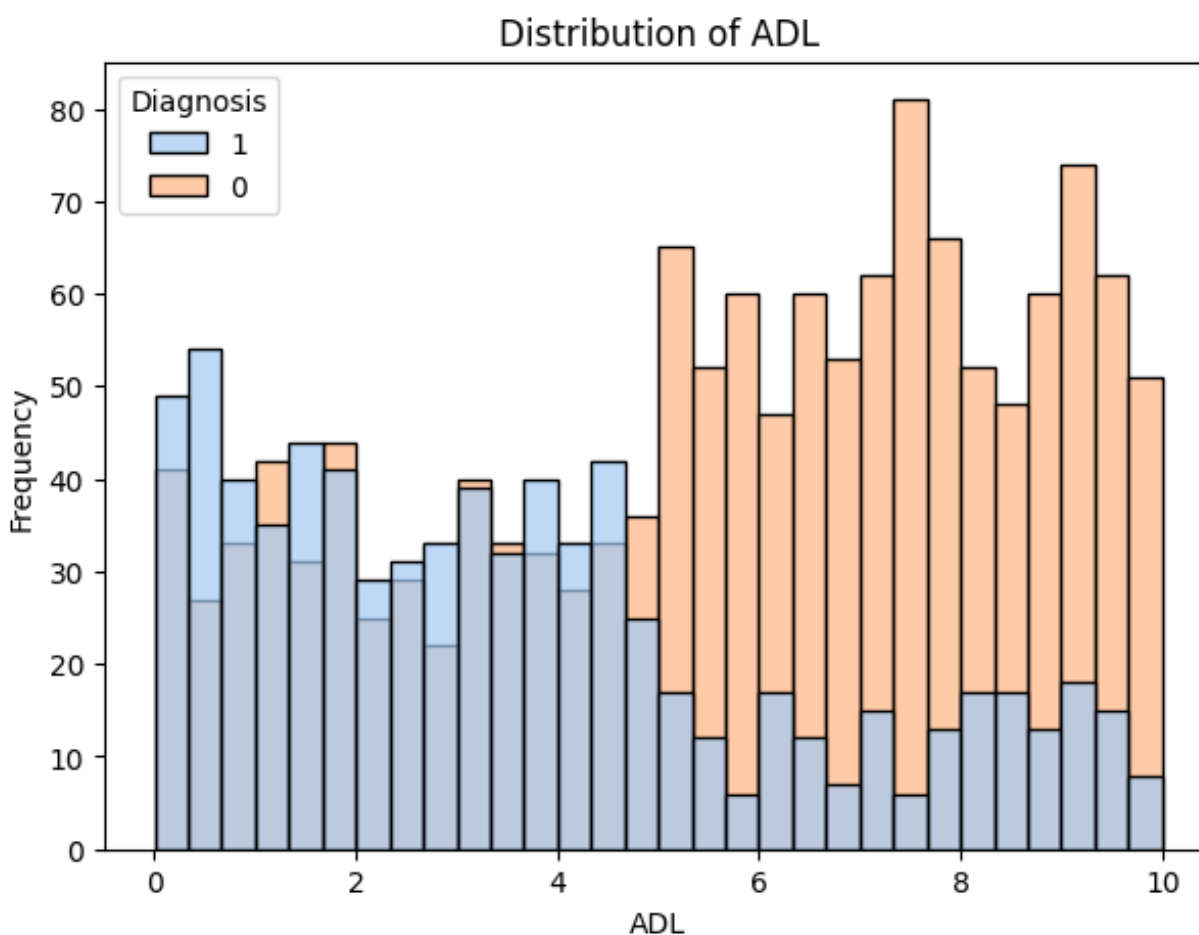


Figure 4: The histogram to the left shows the distribution of Activities of Daily Living score. Similar to Figures 2 and 3, Lower scores indicate greater impairment. The distribution of the positive case has a right skew while the distribution of the negative case has a left skew showing more positively diagnosed patients show greater impairment.

Figure 5

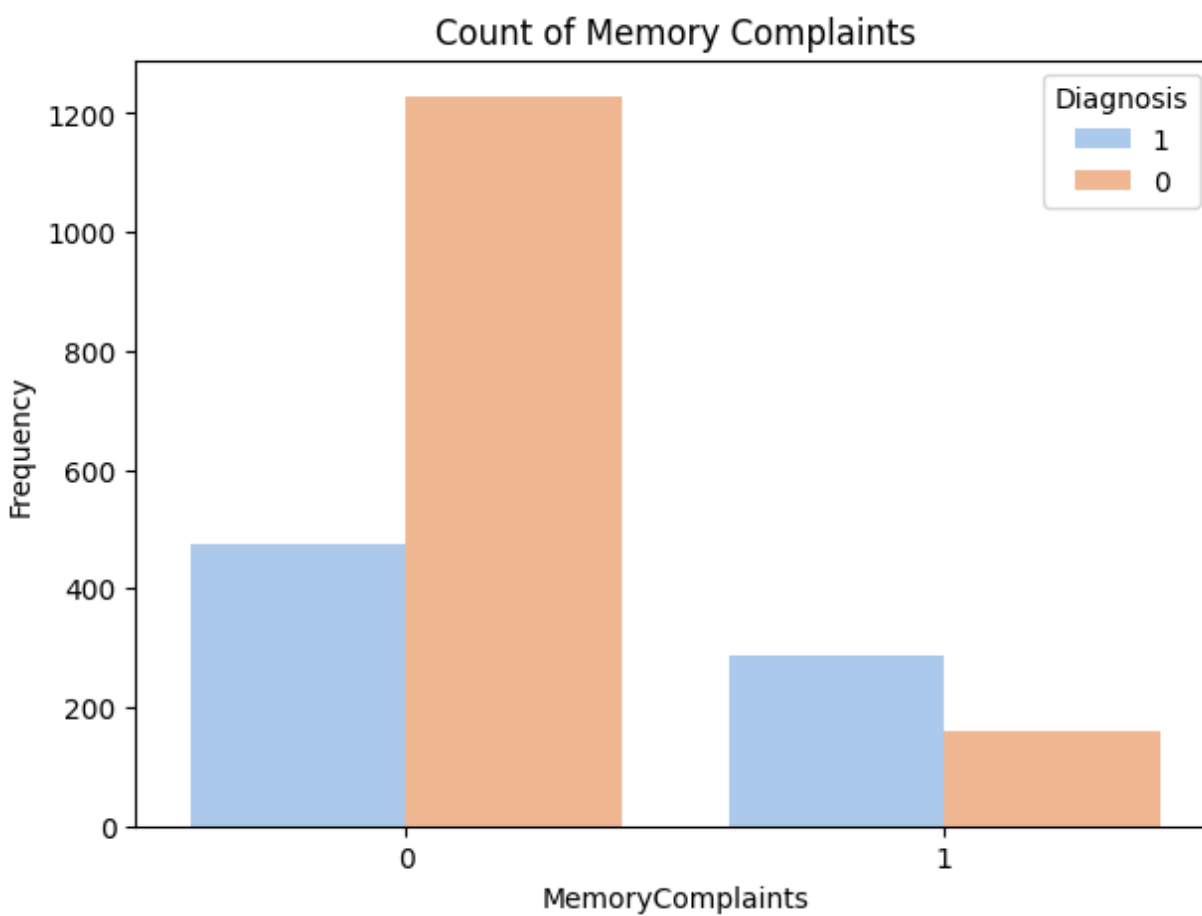


Figure 5: In the countplot to the left, we can see more patients diagnosed with Alzheimer's reported memory complaints than patients without.

Figure 6

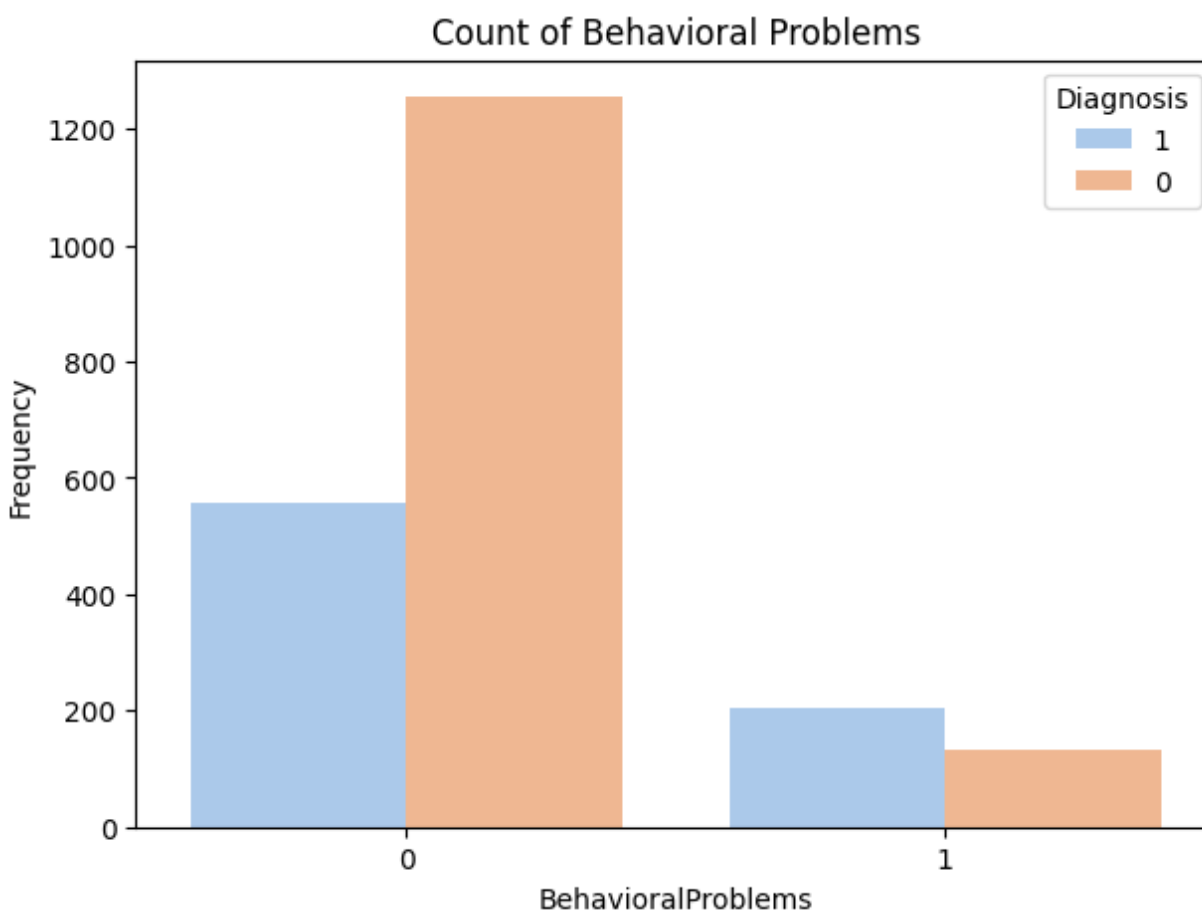


Figure 6: In the countplot to the left, we can see more patients diagnosed with Alzheimer's reported behavioral problems than patients without.

Modeling

After identifying some notable trends and correlations in our dataset, we decided to build a simple Random Forest model with 1000 estimators and a max depth of 5 to extract feature importances. By doing this we were able to identify the features that had the most impact on predicting if a patient was diagnosed with Alzheimer's. Our results can be found in Figure 7 below.

Figure 7

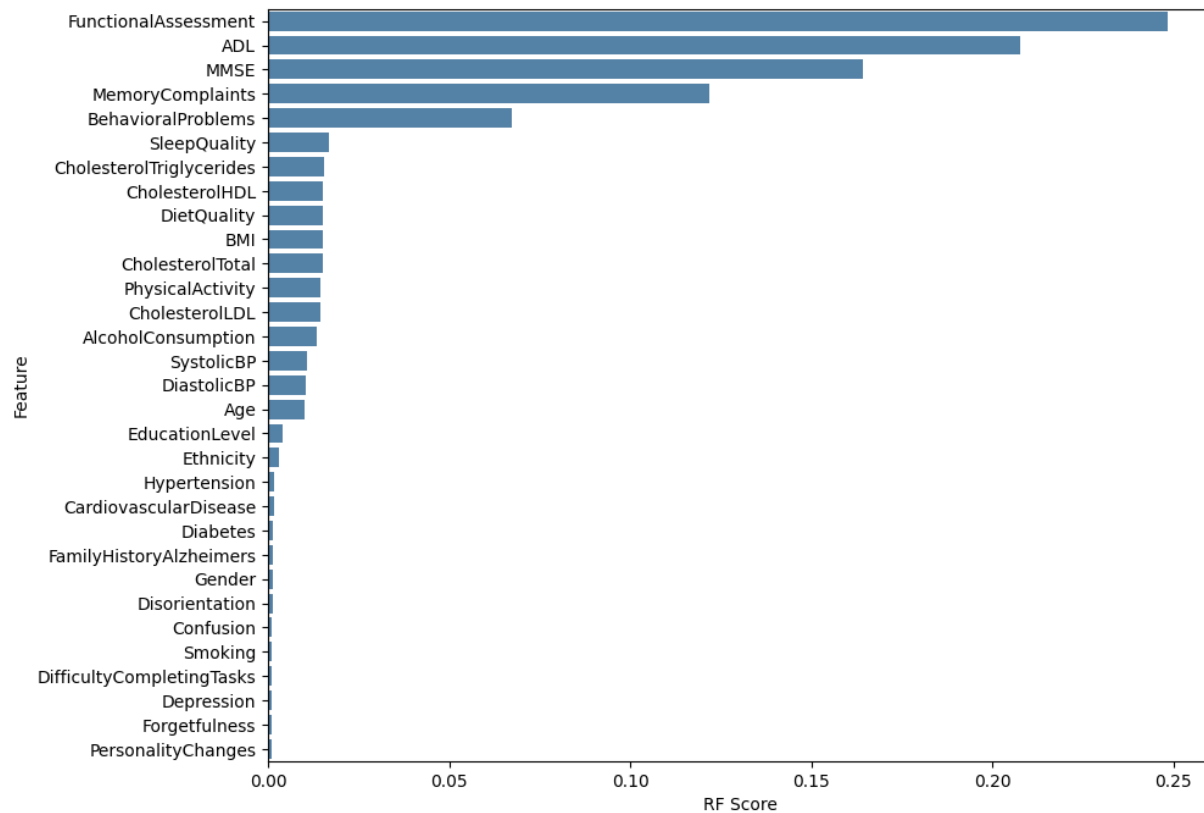


Figure 7: The above chart shows the Random Forest importances ordered by impact to the model. Notice the top 5 important features to the model were the variables we covered in our exploratory analysis. This shows our initial assumptions looking at the charts on what was impactful on identifying Alzheimer's was correct.

Using this model, we were able to achieve a 89% accuracy score over 5 folds cross validation to predict Alzheimer's diagnosis.

Conclusion

In this report, we set out to identify patient attributes that are most indicative of Alzheimer's disease, with the goal of helping flag patients who may be at risk and offering early intervention. Through our Exploratory Data Analysis, we observed several patterns and trends among the dataset, lifestyle factors and their relationship with Alzheimer's diagnosis in particular, which did not show much of a correlation. Factors such as alcohol consumption, physical activity levels, sleep quality levels and diet quality showed similar distributions among positive and negatively diagnosed patients. Which suggests in this data we could not determine if they made a key impact in predicting the likelihood of developing the disease.

We also identified important attributes that patients with Alzheimer's disease reported have based on the data. Those attributes were memory complaints and behavioral problems. We also found the most important predictors for Alzheimer's were the mental and physical exams each patient took, each showed greater impairment for the positively diagnosed patients. Identifying these key attributes could help future patients be diagnosed sooner.

Further Study

In the future we could look at more interactions between variables to see if we could use feature engineering techniques to create new variables that would be effective in predicting Alzheimer's.

Another thing I would like to look at is more models. We achieved a high accuracy score with a very simple random forest model. It would be interesting to see how high we could get the accuracy score with a more in-depth approach to modeling.

References

Dataset:

<https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset/data>¹

Alzheimer's background information:

https://www.alz.org/alzheimer_s_dementia²